■ Automated Aquaponics System Diagnostic Report

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1. Real-time Environmental Telemetry Matrix

System Parameter	Current Simulation Value	Operational Target Status
Water Temperature	25.93 °C	Optimal: 23 - 28 °C
Water pH Level	6.62	Optimal: 6.5 - 7.5
Dissolved Oxygen (O2)	8.42 mg/L	Optimal: > 5.5 mg/L
Atmospheric Humidity	64.92 %	Optimal: 40 - 80 %
Water Tank Level	33.60 %	Optimal: > 50 %
Ammonia (NH3) Load	0.35 ppm	Optimal: < 0.25 ppm
Nitrate (NO3) Level	14.80 ppm	Optimal: 10 - 40 ppm
Hydro-Pump Flow Rate	0.72 L/min	Optimal: > 1.0 L/min

2. Autonomous Root-Cause & Actionable Diagnostics

■ **[BIOCHEMICAL TOXICITY] Elevated Ammonia Load:** 0.35 ppm

• The Cause: Over-feeding of feedstock, unconsumed feed decomposition, or biofilter bacteria breakdown.

• Immediate Action Required: Execute an automated 20% water flush, restrict feeding calculator access for 12 cycles, and verify biofilter mesh status.

■ **[HYDRO-SHORTAGE] Low Water Tank Level:** 33.60 %

• The Cause: Physical pipe rupture, evaporation rate leakage, or pump backflow.

• Immediate Action Required: Trigger the electronic Solenoid valve to initiate auto-refill sequence.

3. Fish Biomass Configuration Status

• Total Fish Count: **250 Pcs** | Net Calculated Biomass: **58.75 kg**

• Global Ecosystem Health Rating: **100/100** | Feed Efficiency Score: **70/100**